Proof that $E[X] = \int_0^\infty [1 - F_X(x)]\, dx$

Theorem 1. *Let X be a non-negative random variable with cumulative distribution function $F_X(x) = P(X \le x)$. Then*

$$E[X] = \int_0^\infty \big[1 - F_X(x)\big]\, dx = \int_0^\infty P(X > x)\, dx.$$

Proof. We give two proofs: one using Fubini's theorem, and one using integration by parts.

Proof 1 (Fubini's Theorem).

For any non-negative random variable X, we can write

$$X = \int_0^\infty \mathbf{1}_{\{X>x\}}\, dx,$$

where $\mathbf{1}_{\{X>x\}}$ is the indicator function that equals 1 when $X > x$ and 0 otherwise. This identity holds because for a fixed outcome ω, the integrand equals 1 for $x \in [0, X(\omega))$ and 0 for $x \ge X(\omega)$, so the integral evaluates to $X(\omega)$.

Taking expectations of both sides:

$$E[X] = E\left[\int_0^\infty \mathbf{1}_{\{X>x\}}\, dx\right].$$

Since $\mathbf{1}_{\{X>x\}} \ge 0$, we may apply Tonelli's theorem (the non-negative version of Fubini's theorem) to interchange the expectation and the integral:

$$E[X] = \int_0^\infty E\big[\mathbf{1}_{\{X>x\}}\big]\, dx = \int_0^\infty P(X > x)\, dx = \int_0^\infty \big[1 - F_X(x)\big]\, dx.$$

This completes the proof. Note that the result holds whether $E[X]$ is finite or infinite.

Proof 2 (Integration by Parts).

Assume X is a non-negative continuous random variable with density f_X. By definition,

$$E[X] = \int_0^\infty x\, f_X(x)\, dx.$$

We apply integration by parts with

$$u = x, \qquad dv = f_X(x)\, dx,$$

so that

$$du = dx, \qquad v = F_X(x).$$

Then

$$E[X] = \Big[x\, F_X(x)\Big]_0^\infty - \int_0^\infty F_X(x)\, dx.$$

For the boundary term, note that $0 \cdot F_X(0) = 0$. For the upper limit, we use the substitution $x\,F_X(x) = x - x[1 - F_X(x)] = x - x\,P(X > x)$. Since $E[X] < \infty$ implies $x\,P(X > x) \to 0$ as $x \to \infty$ (by Markov's inequality, $P(X > x) \le E[X]/x$), we get

$$\lim_{x\to\infty} x\,F_X(x) = \lim_{x\to\infty} x = \infty \quad \text{formally,}$$

but more carefully:

$$\lim_{M\to\infty} \left[M\,F_X(M) - \int_0^M F_X(x)\,dx \right] = \lim_{M\to\infty} \int_0^M \bigl[1 - F_X(x)\bigr]\,dx - M\bigl[1 - F_X(M)\bigr].$$

It is cleaner to proceed directly. Write $f_X(x) = -\frac{d}{dx}[1 - F_X(x)]$ and integrate by parts with $u = x$ and $dv = -d[1 - F_X(x)]$:

$$\begin{aligned} E[X] &= \int_0^\infty x\,f_X(x)\,dx \\ &= \Bigl[-x\bigl(1 - F_X(x)\bigr)\Bigr]_0^\infty + \int_0^\infty \bigl[1 - F_X(x)\bigr]\,dx \\ &= 0 + \int_0^\infty \bigl[1 - F_X(x)\bigr]\,dx, \end{aligned}$$

where the boundary term vanishes because $1 - F_X(0^-) \le 1$ gives $0 \cdot 1 = 0$ at the lower limit, and $x[1 - F_X(x)] \le x\,P(X > x) \to 0$ at the upper limit (by Markov's inequality when $E[X] < \infty$). Hence

$$\boxed{E[X] = \int_0^\infty \bigl[1 - F_X(x)\bigr]\,dx.}$$

□